# Proving an isomorphism step by step

In this document I show how to prove an isomorphism on paper, by constructing the proof trees of the functions and proving that they are inverses of each other using the rules of the language.

## 1. Language definition

We are working with the `Finite` language, which consists of these types:

$$
\begin{aligned}
_ \Rightarrow _ &: \mathrm{Ty} \to \mathrm{Ty} \to \mathrm{Ty} \\
\bot &: \mathrm{Ty} \\
_ + _ &: \mathrm{Ty} \to \mathrm{Ty} \to \mathrm{Ty} \\
\top &: \mathrm{Ty} \\
_ \times _ &: \mathrm{Ty} \to \mathrm{Ty} \to \mathrm{Ty}
\end{aligned}
$$

The language has the following constructors and iterators, written as proof rules. Note that for case we use the argument order of the lecture notes, not the practices. We also include the logic's hyp rule here.

hyp
$$\frac{}{x : \mathrm{Tm}\, A \vdash x : \mathrm{Tm}\, A}$$

lam
$$\frac{x : \mathrm{Tm}\, A \vdash f : \mathrm{Tm}\, B}{\mathrm{lam}\, \lambda x.f : \mathrm{Tm}\ (A \Rightarrow B)}$$

_ · _
$$\frac{f : Tm(A \Rightarrow B) \qquad x : TmA}{f \cdot x : TmB}$$

exfalso
$$\frac{b : \mathrm{Tm} \bot}{\mathrm{exfalso}\, b : \mathrm{Tm}\, A}$$

inl
$$\frac{a : \mathrm{Tm}\, A}{\mathrm{inl}\, a : \mathrm{Tm}\ (A + B)}$$

inr
$$\frac{b : \mathrm{Tm}\, B}{\mathrm{inr}\, b : \mathrm{Tm}\ (A + B)}$$

case
$$\frac{a : \mathrm{Tm}\, A \vdash f : \mathrm{Tm}\, C \qquad b : \mathrm{Tm}\, B \vdash g : \mathrm{Tm}\, C \qquad ab : \mathrm{Tm}\ (A + B)}{\mathrm{case}\ (\lambda a.f)\ (\lambda b.g)\ ab : \mathrm{Tm}\, C}$$

tt
$$\frac{}{\mathrm{tt} : \mathrm{Tm} \top}$$

_, _
$$\frac{a : \mathrm{Tm}\, A \qquad b : \mathrm{Tm}\, B}{a, b : \mathrm{Tm}\ (A \times B)}$$

fst
$$\frac{ab : \mathrm{Tm}\ (A \times B)}{\mathrm{fst}\, ab : \mathrm{Tm}\, A}$$

snd
$$\frac{ab : \mathrm{Tm}\ (A \times B)}{\mathrm{snd}\, ab : \mathrm{Tm}\, B}$$

The following $\beta$ and $\eta$ rules define how the above types and iterators work:

$$
\begin{aligned}
\Rightarrow\beta: &\quad \mathrm{lam}\, b \cdot a = b\ a \\
+\beta_1: &\quad \mathrm{case}\, f g (\mathrm{inl}\, a) = f\ a \\
+\beta_2: &\quad \mathrm{case}\, f g (\mathrm{inr}\, b) = g\ b \\
\times\beta_1: &\quad \mathrm{fst}\ (a,\ b) = a \\
\times\beta_2: &\quad \mathrm{snd}\ (a,\ b) = b \\
\Rightarrow\eta: &\quad (f : Tm(A \Rightarrow B)) = \mathrm{lam}\, \lambda x.f \cdot x \\
\bot\eta: &\quad (t : \mathrm{Tm}\, \bot \to \mathrm{Tm}\, A) = \mathrm{exfalso} \\
+\eta: &\quad (t : \mathrm{Tm}\ (A + B) \to \mathrm{Tm}\, C) = \mathrm{case}\ (\lambda a.t\ (\mathrm{inl}\, a))\ (\lambda b.t\ (\mathrm{inr}\, b)) \\
\top\eta: &\quad (t : \mathrm{Tm}\, \top) = \mathrm{tt} \\
\times\eta: &\quad (t : \mathrm{Tm}\ (A \times B)) = (\mathrm{fst}\, t,\ \mathrm{snd}\, t)
\end{aligned}
$$

## 2. Proving an isomorphism

Let us prove the following isomorphism between regular pairs, and ones encoded using booleans (represented here by $\top + \top$):

$$\top + \top \Rightarrow A \cong A \times A$$

### 2.1. Function from left to right

First, we must provide the function from the type on the left to the one on the right, which has the type $\mathrm{Tm}\ (\top + \top \Rightarrow A) \to \mathrm{Tm}\ (A \times A)$. We do this by constructing a proof tree. I show every step, and the rules used between each step. On the exam, you do not need to do this, just use a similar reasoning to reach the final term. I use $\square$ to denote holes where terms will go, and $\bigcirc$ for types. You can use any of the placeholders discussed on the practices, including empty space, underlines, placeholder letters written in pencil (make sure everything is written in pen before you submit!), etc.

Due to fact that the tree is very wide, it is placed on the next, extra wide page.

### 2.2. Function from right to left

Next, we must define the function in the opposite direction, from the type on the right to the one on the left, with the type $\mathrm{Tm}\ (A \times A) \to \mathrm{Tm}\ (\top + \top \Rightarrow A)$. I follow the same practice here as I did for the previous one.

This tree is similarly wide, so it can be found after the other tree, on its own extra wide page. The rest of the proof can be found after the trees.

$$f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \Box : \mathrm{Tm}\ (A \times A)$$

$$\downarrow _, _$$

$$\frac{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \Box_1 : \mathrm{Tm}\, A \qquad f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \Box_2 : \mathrm{Tm}\, A}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \Box_1, \Box_2 : \mathrm{Tm}\ (A \times A)}$$

$$\downarrow _ \cdot _ \text{ (twice)}$$

$$\frac{\dfrac{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \Box_1 : \mathrm{Tm} \bigcirc_1 \Rightarrow \mathrm{Tm}\, A \qquad f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \Box_3 : \mathrm{Tm} \bigcirc_1}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \Box_1 \cdot \Box_3 : \mathrm{Tm}\, A} \qquad \dfrac{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \Box_2 : \mathrm{Tm} \bigcirc_2 \Rightarrow \mathrm{Tm}\, A \qquad f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \Box_4 : \mathrm{Tm} \bigcirc_2}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \Box_2 \cdot \Box_4 : \mathrm{Tm}\, A}}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \Box_1 \cdot \Box_3, \Box_2 \cdot \Box_4 : \mathrm{Tm}\ (A \times A)}$$

$$\downarrow \text{hyp} \text{ (twice)}$$

$$\frac{\dfrac{\dfrac{\checkmark}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash f : \mathrm{Tm}\ (\top + \top) \Rightarrow \mathrm{Tm}\, A} \qquad f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \Box_3 : \mathrm{Tm}\ (\top + \top)}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash f \cdot \Box_3 : \mathrm{Tm}\, A} \qquad \dfrac{\dfrac{\checkmark}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash f : \mathrm{Tm}\ (\top + \top) \Rightarrow \mathrm{Tm}\, A} \qquad f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \Box_4 : \mathrm{Tm}\ (\top + \top)}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash f \cdot \Box_4 : \mathrm{Tm}\, A}}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash f \cdot \Box_3, f \cdot \Box_4 : \mathrm{Tm}\ (A \times A)}$$

$$\downarrow \mathrm{inl}, \mathrm{inr}$$

$$\frac{\dfrac{\dfrac{\checkmark}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash f : \mathrm{Tm}\ (\top + \top) \Rightarrow \mathrm{Tm}\, A} \qquad \dfrac{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \Box_3 : \mathrm{Tm}\, \top}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \mathrm{inl}\, \Box_3 : \mathrm{Tm}\ (\top + \top)}}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash f \cdot \mathrm{inl}\, \Box_3 : \mathrm{Tm}\, A} \qquad \dfrac{\dfrac{\checkmark}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash f : \mathrm{Tm}\ (\top + \top) \Rightarrow \mathrm{Tm}\, A} \qquad \dfrac{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \Box_4 : \mathrm{Tm}\, \top}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \mathrm{inr}\, \Box_4 : \mathrm{Tm}\ (\top + \top)}}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash f \cdot \mathrm{inr}\, \Box_4 : \mathrm{Tm}\, A}}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash f \cdot \mathrm{inl}\, \Box_3, f \cdot \mathrm{inr}\, \Box_4 : \mathrm{Tm}\ (A \times A)}$$

$$\downarrow \mathrm{tt} \text{ (twice)}$$

$$\frac{\dfrac{\dfrac{\checkmark}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash f : \mathrm{Tm}\ (\top + \top) \Rightarrow \mathrm{Tm}\, A} \qquad \dfrac{\dfrac{\checkmark}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \mathrm{tt} : \mathrm{Tm}\, \top}}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \mathrm{inl}\, \mathrm{tt} : \mathrm{Tm}\ (\top + \top)}}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash f \cdot \mathrm{inl}\, \mathrm{tt} : \mathrm{Tm}\, A} \qquad \dfrac{\dfrac{\checkmark}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash f : \mathrm{Tm}\ (\top + \top) \Rightarrow \mathrm{Tm}\, A} \qquad \dfrac{\dfrac{\checkmark}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \mathrm{tt} : \mathrm{Tm}\, \top}}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash \mathrm{inr}\, \mathrm{tt} : \mathrm{Tm}\ (\top + \top)}}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash f \cdot \mathrm{inr}\, \mathrm{tt} : \mathrm{Tm}\, A}}{f : \mathrm{Tm}\ (\top + \top \Rightarrow A) \vdash f \cdot \mathrm{inl}\, \mathrm{tt}, f \cdot \mathrm{inr}\, \mathrm{tt} : \mathrm{Tm}\ (A \times A)}$$

$$aa : \mathrm{Tm}\ (A \times A) \vdash \square : \mathrm{Tm}\ (\top + \top \Rightarrow A)$$

$$\Big\downarrow \mathrm{lam}$$

$$\frac{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top) \vdash \square : \mathrm{Tm}\ A}{aa : \mathrm{Tm}\ (A \times A) \vdash \mathrm{lam}\ \lambda x.\square : \mathrm{Tm}\ (\top + \top \Rightarrow A)}$$

$$\Big\downarrow \mathrm{case}$$

$$\frac{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top), a : \mathrm{Tm}\ \bigcirc_1 \vdash \square_1 : \mathrm{Tm}\ A \qquad aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top), b : \mathrm{Tm}\ \bigcirc_2 \vdash \square_2 : \mathrm{Tm}\ A \qquad aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top) \vdash \square_3 : \mathrm{Tm}\ (\bigcirc_1 + \bigcirc_2)}{\dfrac{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top) \vdash \mathrm{case}\ (\lambda a.\square_1)\ (\lambda b.\square_2)\ \square_3 : \mathrm{Tm}\ A}{aa : \mathrm{Tm}\ (A \times A) \vdash \mathrm{lam}\ \lambda x.\, \mathrm{case}\ (\lambda a.\square_1)\ (\lambda b.\square_2)\ \square_3 : \mathrm{Tm}\ (\top + \top \Rightarrow A)}}$$

$$\Big\downarrow \mathrm{hyp}$$

$$\frac{\dfrac{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top), a : \mathrm{Tm}\ \top \vdash \square_1 : \mathrm{Tm}\ A \qquad aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top), b : \mathrm{Tm}\ \top \vdash \square_2 : \mathrm{Tm}\ A \qquad \dfrac{\checkmark}{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top) \vdash x : \mathrm{Tm}\ (\top + \top)}}{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top) \vdash \mathrm{case}\ (\lambda a.\square_1)\ (\lambda b.\square_2)\ x : \mathrm{Tm}\ A}}{aa : \mathrm{Tm}\ (A \times A) \vdash \mathrm{lam}\ \lambda x.\, \mathrm{case}\ (\lambda a.\square_1)\ (\lambda b.\square_2)\ x : \mathrm{Tm}\ (\top + \top \Rightarrow A)}$$

$$\Big\downarrow \mathrm{fst}, \mathrm{snd}$$

$$\frac{\dfrac{\dfrac{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top), a : \mathrm{Tm}\ \top \vdash \square_1 : \mathrm{Tm}\ (A \times \bigcirc_1)}{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top), a : \mathrm{Tm}\ \top \vdash \mathrm{fst}\ \square_1 : \mathrm{Tm}\ A} \qquad \dfrac{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top), b : \mathrm{Tm}\ \top \vdash \square_2 : \mathrm{Tm}\ (\bigcirc_2 \times A)}{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top), b : \mathrm{Tm}\ \top \vdash \mathrm{snd}\ \square_2 : \mathrm{Tm}\ A} \qquad \dfrac{\checkmark}{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top) \vdash x : \mathrm{Tm}\ (\top + \top)}}{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top) \vdash \mathrm{case}\ (\lambda a.\, \mathrm{fst}\ \square_1)\ (\lambda b.\, \mathrm{snd}\ \square_2)\ x : \mathrm{Tm}\ A}}{aa : \mathrm{Tm}\ (A \times A) \vdash \mathrm{lam}\ \lambda x.\, \mathrm{case}\ (\lambda a.\, \mathrm{fst}\ \square_1)\ (\lambda b.\, \mathrm{snd}\ \square_2)\ x : \mathrm{Tm}\ (\top + \top \Rightarrow A)}$$

$$\Big\downarrow \mathrm{hyp}\ \ (\mathrm{twice})$$

$$\frac{\dfrac{\dfrac{\dfrac{\checkmark}{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top), a : \mathrm{Tm}\ \top \vdash aa : \mathrm{Tm}\ (A \times A)}}{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top), a : \mathrm{Tm}\ \top \vdash \mathrm{fst}\ aa : \mathrm{Tm}\ A} \qquad \dfrac{\dfrac{\checkmark}{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top), b : \mathrm{Tm}\ \top \vdash aa : \mathrm{Tm}\ (A \times A)}}{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top), b : \mathrm{Tm}\ \top \vdash \mathrm{snd}\ aa : \mathrm{Tm}\ A} \qquad \dfrac{\checkmark}{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top) \vdash x : \mathrm{Tm}\ (\top + \top)}}{aa : \mathrm{Tm}\ (A \times A), x : \mathrm{Tm}\ (\top + \top) \vdash \mathrm{case}\ (\lambda a.\, \mathrm{fst}\ aa)\ (\lambda b.\, \mathrm{snd}\ aa)\ x : \mathrm{Tm}\ A}}{aa : \mathrm{Tm}\ (A \times A) \vdash \mathrm{lam}\ \lambda x.\, \mathrm{case}\ (\lambda a.\, \mathrm{fst}\ aa)\ (\lambda b.\, \mathrm{snd}\ aa)\ x : \mathrm{Tm}\ (\top + \top \Rightarrow A)}$$

### 2.3. Left to right to left inverse

Next we must prove that if we go from the left side to the right and then back, we reach the same value we started with. For this we use the two terms we created above with the trees, and present a step-by-step proof. This is similar to what you would write at the exam as well. Our input is represented by the variable $f$.

$$
\begin{aligned}
\text{lam}\,\lambda x.\,\text{case}\ (\lambda a.\underline{\text{fst}(f\cdot\text{inl}\,\text{tt}, f\cdot\text{inr}\,\text{tt})})\ (\lambda b.\,\text{snd}(f\cdot\text{inl}\,\text{tt}, f\cdot\text{inr}\,\text{tt}))\ x &\overset{\times\beta_1}{=}\\
\text{lam}\,\lambda x.\,\text{case}\ (\lambda a.f\cdot\text{inl}\,\text{tt})\ (\lambda b.\underline{\text{snd}(f\cdot\text{inl}\,\text{tt}, f\cdot\text{inr}\,\text{tt})})\ x &\overset{\times\beta_2}{=}\\
\text{lam}\,\lambda x.\,\text{case}\ (\lambda a.f\cdot\text{inl}\,\underline{\text{tt}})\ (\lambda b.f\cdot\text{inr}\,\text{tt})\ x &\overset{\top\eta}{=}\\
\text{lam}\,\lambda x.\,\text{case}\ (\lambda a.f\cdot\text{inl}\,a)\ (\lambda b.f\cdot\text{inr}\,\underline{\text{tt}})\ x &\overset{\top\eta}{=}\\
\text{lam}\,\lambda x.\underline{\text{case}\ (\lambda a.f\cdot\text{inl}\,a)\ (\lambda b.f\cdot\text{inr}\,b)\ x} &\overset{+\eta}{=}\\
\underline{\text{lam}\,\lambda x.f\cdot x} &\overset{\Rightarrow\eta}{=}\\
f\ \square
\end{aligned}
$$

### 2.4. Right to left to right inverse

Finally, we prove that going from right to left to right also results in the original input. We use the same method as above, except this time the input is $aa$.

$$
\begin{aligned}
\underline{(\text{lam}\,\lambda x.\,\text{case}\ (\lambda a.\,\text{fst}\,aa)\ (\lambda b.\,\text{snd}\,aa)\ x)\cdot\text{inl}\,\text{tt}}, (\text{lam}\,\lambda x.\,\text{case}\ (\lambda a.\,\text{fst}\,aa)\ (\lambda b.\,\text{snd}\,aa)\ x)\cdot\text{inr}\,\text{tt} &\overset{\Rightarrow\beta}{=}\\
\text{case}\ (\lambda a.\,\text{fst}\,aa)\ (\lambda b.\,\text{snd}\,aa)\ (\text{inl}\,\text{tt}), \underline{(\text{lam}\,\lambda x.\,\text{case}\ (\lambda a.\,\text{fst}\,aa)\ (\lambda b.\,\text{snd}\,aa)\ x)\cdot\text{inr}\,\text{tt}} &\overset{\Rightarrow\beta}{=}\\
\underline{\text{case}\ (\lambda a.\,\text{fst}\,aa)\ (\lambda b.\,\text{snd}\,aa)\ (\text{inl}\,\text{tt})}, \text{case}\ (\lambda a.\,\text{fst}\,aa)\ (\lambda b.\,\text{snd}\,aa)\ (\text{inr}\,\text{tt}) &\overset{+\beta_1}{=}\\
\text{fst}\,aa, \underline{\text{case}\ (\lambda a.\,\text{fst}\,aa)\ (\lambda b.\,\text{snd}\,aa)\ (\text{inr}\,\text{tt})} &\overset{+\beta_2}{=}\\
\underline{\text{fst}\,aa, \text{snd}\,aa} &\overset{\times\eta}{=}\\
aa\ \square
\end{aligned}
$$

With that, we have proven the isomorphism.